

ONYX-01: A Fictional Transforming Robot

Technical Concept Dossier

Prompt Engineering Learning Project

September 2026

Abstract

ONYX-01 connects AI-generated visual design, mathematical modeling, control-flow diagrams, and digital presentation in a single educational prototype. All specifications are fictional educational assumptions. No physical robot or measured performance is claimed.

1 Visual design

The design uses layered matte obsidian armor, graphite mechanical surfaces, a pointed helmet, narrow white-blue eyes, a V-shaped chest, and shoulder wheels. A single reference image anchors subsequent images. Generated details are visual concepts rather than manufacturing drawings.

2 Concept specifications

Parameter	Fictional concept value
Battery energy	2400 Wh (2.4 kWh)
Model speed domain	0–60 km/h
Base power consumption	200 W
Speed coefficient	$0.5 \text{ W}/(\text{km/h})^2$
Low-battery threshold	20%

3 Energy model

Let v denote speed in km/h, E energy in Wh, and P power in W. Assume

$$E = 2400 \text{ Wh}, \tag{1}$$

$$P(v) = 200 + 0.5v^2 \text{ W}, \tag{2}$$

$$t(v) = \frac{E}{P(v)} = \frac{2400}{200 + 0.5v^2} \text{ h}, \quad 0 \leq v \leq 60. \tag{3}$$

The coefficient carries the units stated in the table, so both terms in $P(v)$ are powers. Energy divided by power gives time.

Speed (km/h)	Power (W)	Runtime (h)
0	200	12
20	400	6
40	1000	2.4
60	2000	1.2

4 Control flow

The conceptual controller first reads its sensors. Below 20% battery, it stops the mission and requests charging. Otherwise, it checks for obstacles. An obstacle causes a stop and replanning before another sensor reading. With a clear path, the robot moves and repeats sensing. This is explanatory logic, not a certified controller.

5 Prompt iteration

The first prompt requested a short introduction. The revised prompt specified the compilation engine, document structure, two formatting examples, consistent numerical assumptions, and explicit limitations. This makes the resulting source easier to inspect and compile.

6 Limitations

The energy model ignores terrain, payload, temperature, changing battery efficiency, and transient motor loads. The transformation mechanism is not physically validated. AI-generated images can contain inconsistent geometry.